Multiple Observations in Same Subject

"Repeated Measures"

I. Paired *t* Test
   A. Idea
   B. Test procedure
   C. Examples

II. Repeated Measures ANOVA
   A. Experiment Framework
   B. Partitioning Variability
   C. Procedure

Outline

1. **Multiple Linear Regression (Recap)**
2. In-class exercise : Infant SBP versus Birthweight, Age
3. Logistic Regression
4. Homework 3 Overview and Guidance

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
Interval (and drawn from normally distributed populations*)	Unpaired *t* test (Chapter 4)	Analysis of variance (Chapter 3)	Paired *t* test (Chapter 9)	Repeated-measures analysis of variance (Chapter 9)	Linear regression, Pearson product-moment correlation, or Bland-Altman analysis (Chapter 8)
Nominal	z Test Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5) Logistic regression (slides)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

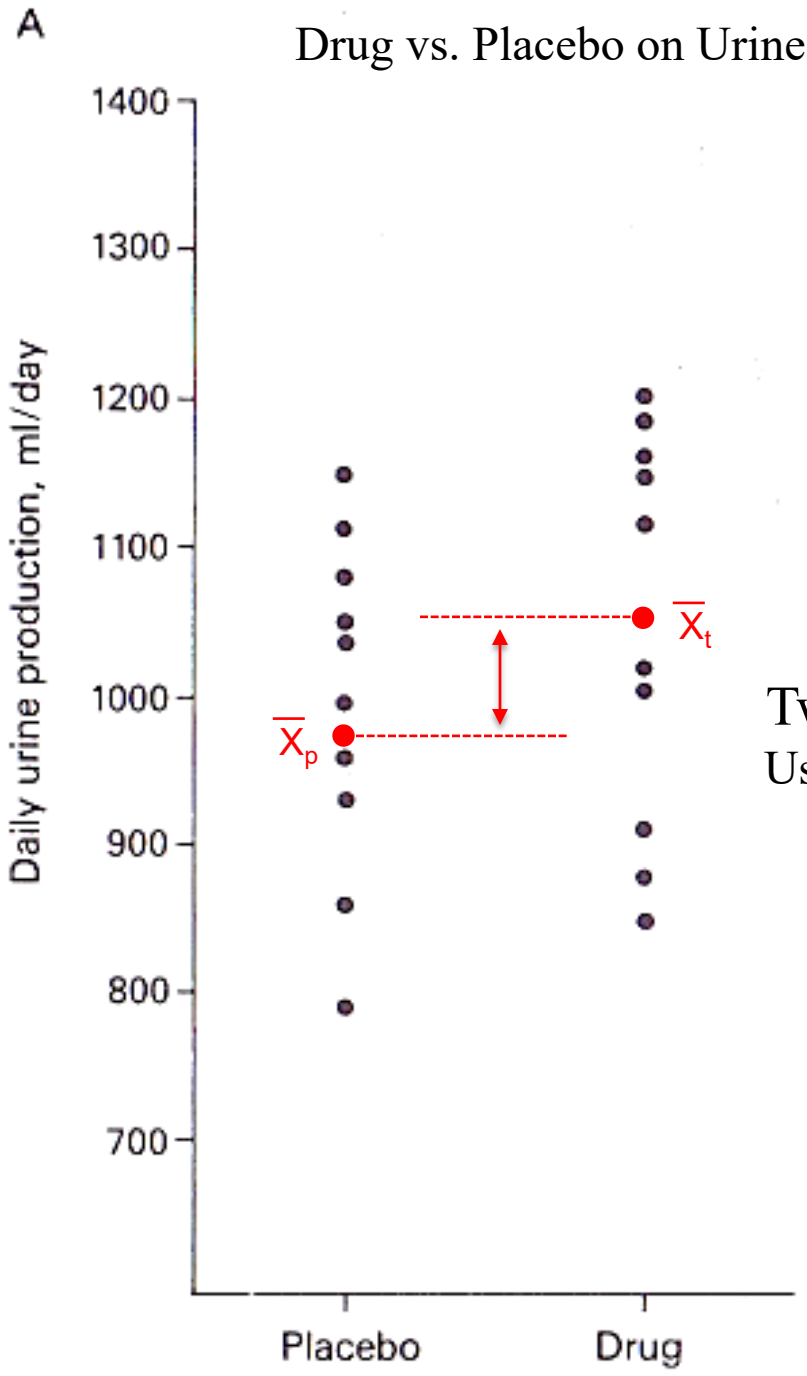
*If the assumption of normally distributed populations is not met, rank the observations and use the methods for data measured on an ordinal scale.

†Not covered in this text.

‡Or interval data that are not necessarily normally distributed.

Unpaired(!) t Test

(what we did before)



$$H_0 : \mu_p = \mu_d$$

Two groups made up of different individuals:
Using (unpaired) t Test, conclude no treatment effect

Unpaired(!) *t* Test

(what we did before)

$$t = \frac{\text{difference in sample means}}{\text{standard error of difference of sample means}}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} \qquad s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

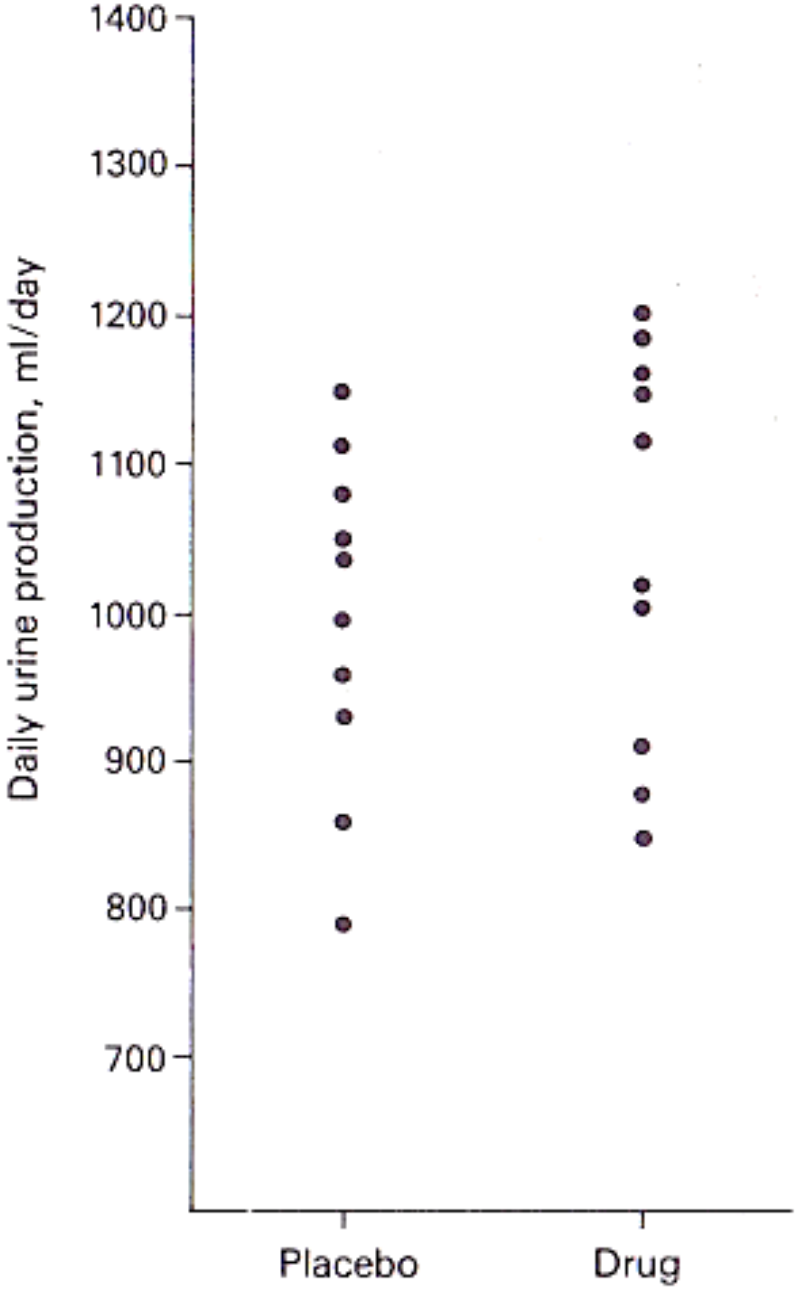
If $|t| > t_\alpha(\nu)$

$$\nu = n_1 + n_2 - 2$$

Reject null hypothesis: $P < \alpha$

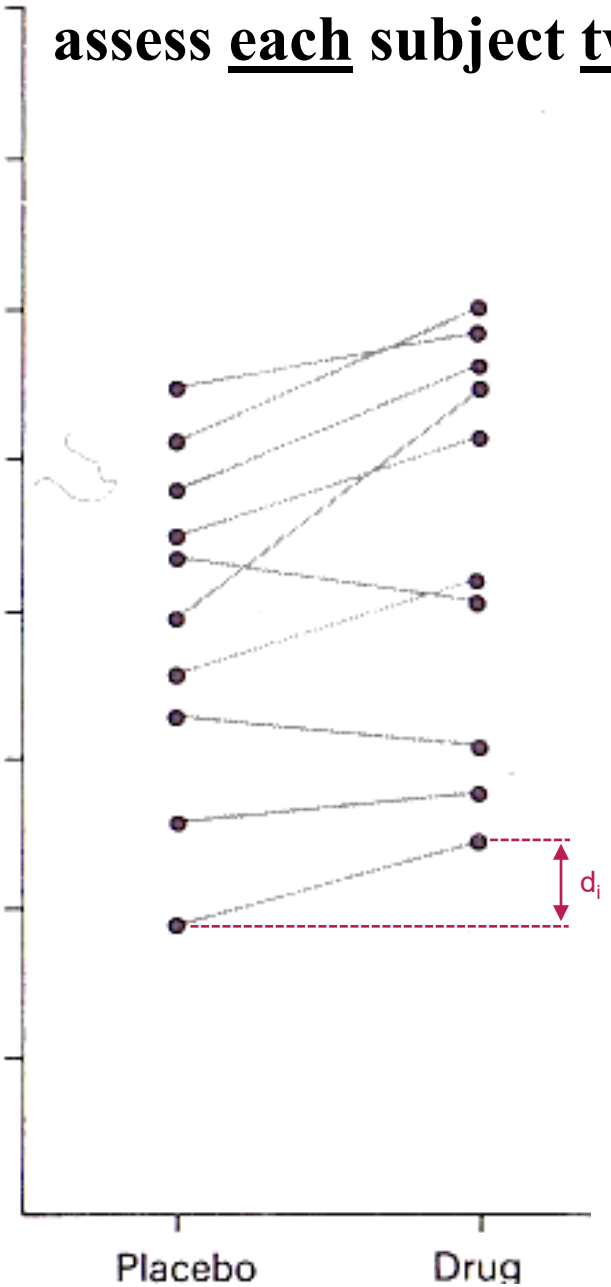
A

Drug vs. Placebo on Urine Production



B

NEW STUDY DESIGN:
assess <u>each</u> subject <u>twice</u>



Paired *t* Test

New Variable – Difference in response for each individual

Treatment 1 response in subject *i*

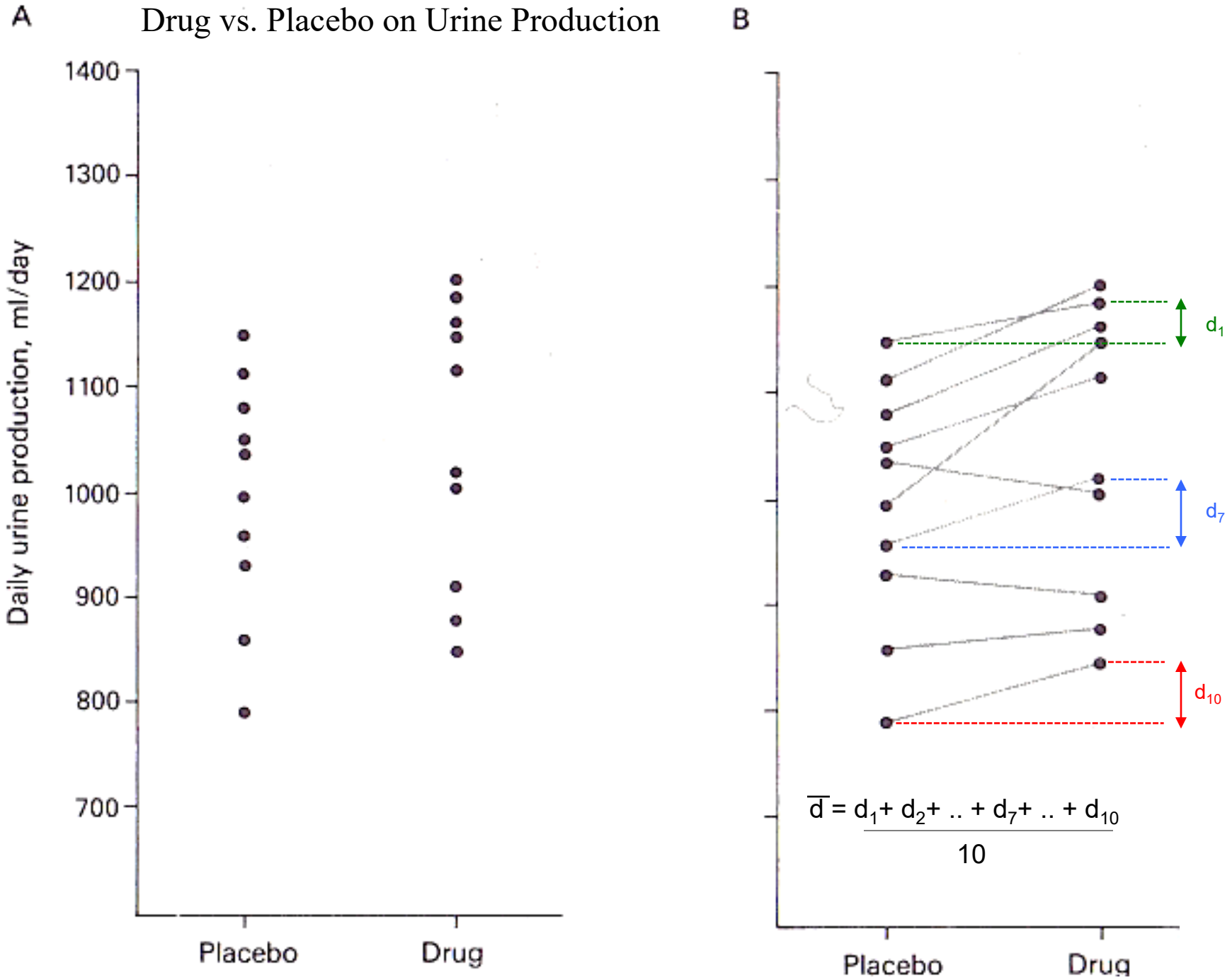
$$d_i = x1_i - x2_i \qquad i = 1, \ldots, n$$

Treatment 2 response in subject *i*

$$t = \frac{\text{Avg. difference} - \text{True value}}{\text{standard error of avg. diff.}} \qquad t = \frac{\bar{d} - \delta}{s_{\bar{d}}}$$

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i \qquad s_d^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(d_i - \bar{d}\right)^2 \qquad s_{\bar{d}} = \frac{s_d}{\sqrt{n}}$$

Drug vs. Placebo on Urine Production



Paired *t* Test

$$t = \frac{\text{Avg. difference} - \text{True value}}{\text{standard error of avg. diff.}}$$

$$t = \frac{\bar{d} - \delta}{s_{\bar{d}}}$$

Null hypothesis: No Difference in Response $H_0 : \delta = 0$

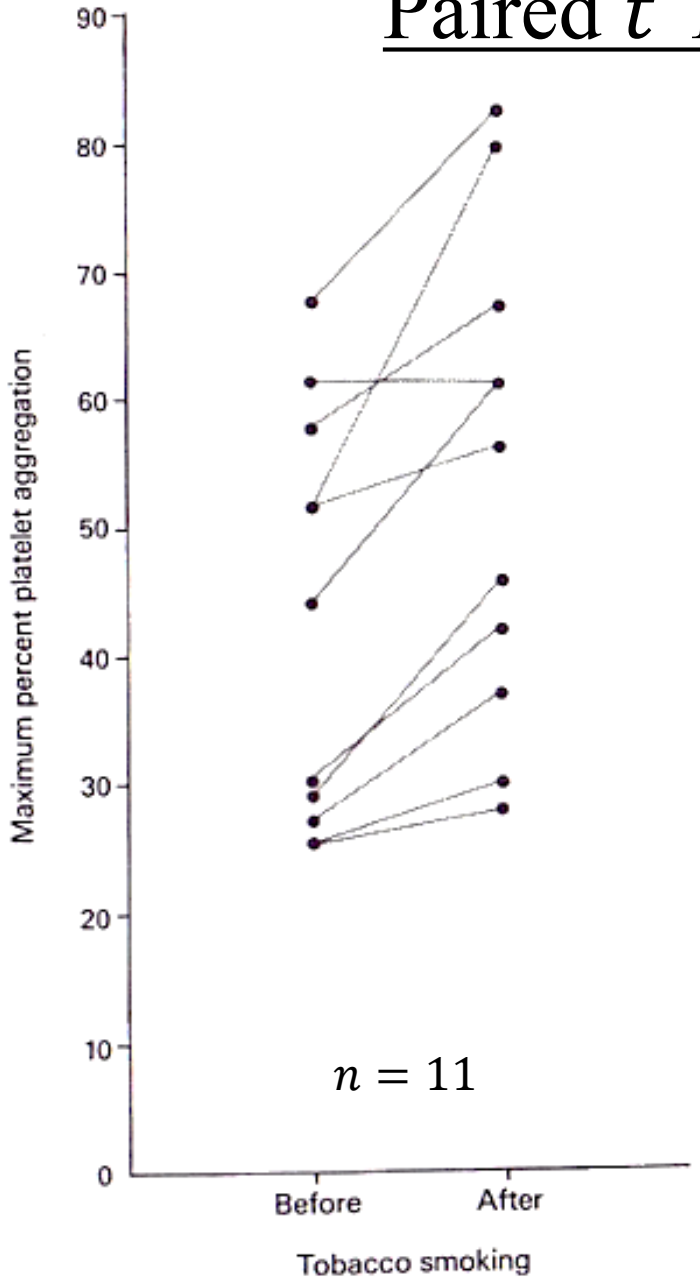
$$t = \frac{\bar{d}}{s_{\bar{d}}} \qquad \nu = n - 1$$

$|t| > t_\alpha(\nu)$ Reject null hypothesis: $P < \alpha$

(using two-tailed t table)

Note: same assumptions are made as with unpaired *t* Test
(i.e., Normal data, interval data, etc.)

Paired t Test - Example



Levine et al., Circulation, 1973.

Change in platelet aggregation before and after smoking a cigarette (%)

$$d_i = (2, 4, 10, 12, 16, 15, 4, 27, 9, -1, 15)$$

$$\bar{d} = 10.3\%$$

$$s_d = 8.0\%$$

$$s_{\bar{d}} = \frac{s_d}{\sqrt{n}} = 2.41\%$$

$$t = \frac{\bar{d}}{s_{\bar{d}}} = \frac{10.3}{2.41} = 4.27 \qquad \nu = n - 1 = 10$$

$$|t| = 4.27 > t_{0.002}(10) = 4.144$$

Reject null hypothesis: $P < 0.002$

Lecture Exercise

Oral contraceptive use and cardiovascular disease

Systolic BP (mmHg)	
w/o OC	w/ OC
115	128
112	115
107	106
119	128
115	122
138	145
126	132
105	109
104	102
115	117

$$n = 10$$

Is there a difference in systolic blood pressure?

Rosner , Fundamentals of Biostats., 2006

Lecture Exercise

Oral contraceptive use and cardiovascular disease

Systolic BP (mmHg)	
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115	128
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$$n = 10$$

Is there a difference in systolic blood pressure?

Change in systolic blood pressure (SBP) with oral contraceptive use (mmHg)

$$d_i = (13, 3, -1, 9, 7, 7, 6, 4, -2, 2)$$

$$\bar{d} = 4.8 \text{ mmHg}$$

$$s_d = 4.566 \text{ mmHg}$$

$$s_{\bar{d}} = \frac{s_d}{\sqrt{n}} = 1.444 \text{ mmHg}$$

$$\nu = n - 1 = 9$$

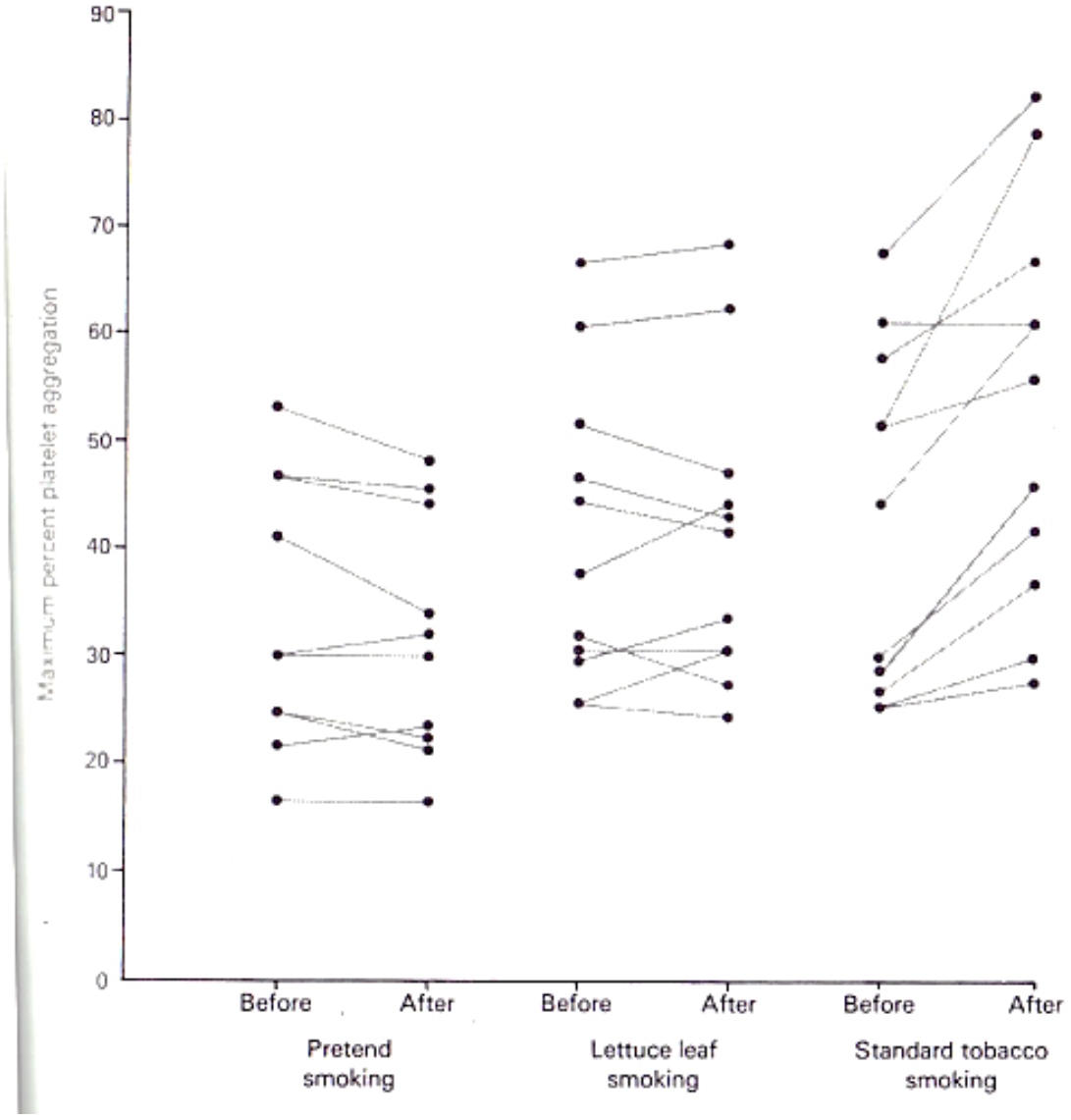
$$|t| = 3.325 > t_{0.01}(9) = 3.25$$

Reject null hypothesis: $P < 0.01$

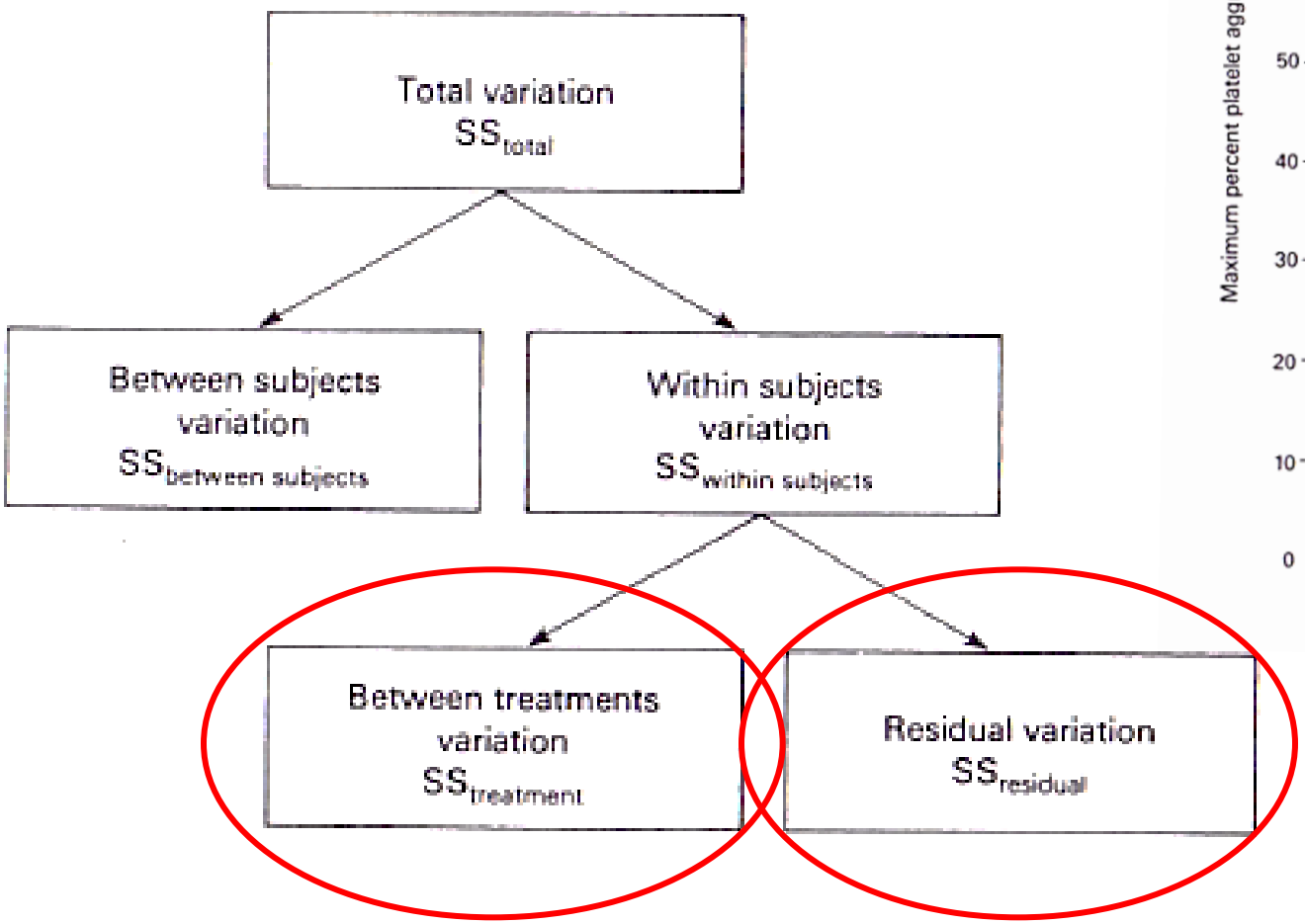
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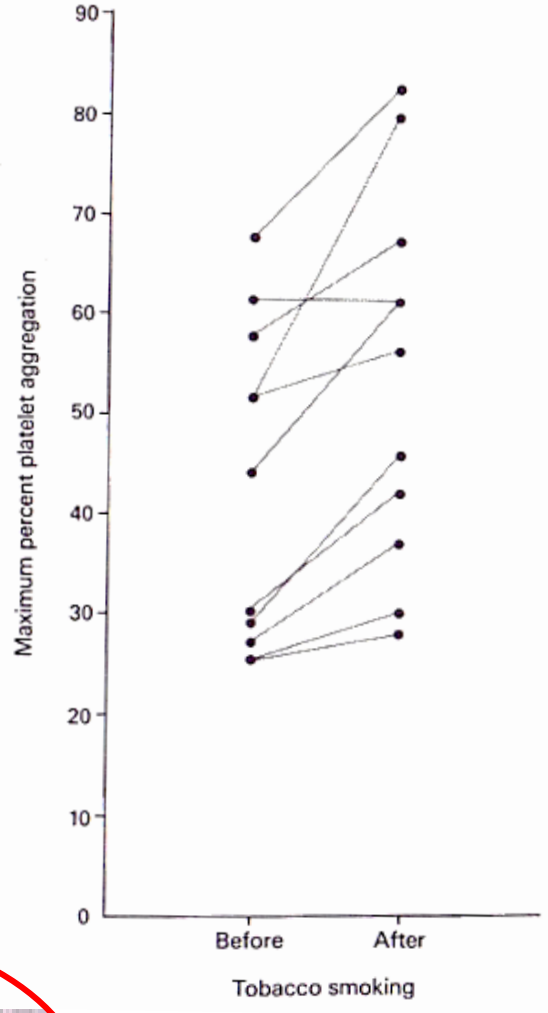
II. Repeated Measures ANOVA

3 Different Paired t Test Experiments to Isolate Effects of Tobacco Smoking



II. Repeated Measures ANOVA





$$F = \frac{MS_{treat}}{MS_{res}}$$

II. Repeated Measures ANOVA
A. Framework

Table 9-4 Notation for Repeated-Measures Analysis of Variance

Experimental subject, $n = 4$	Treatment, $m = 3$			Subject	
	1	2	3	Mean	SS
1	X_{11}	X_{21}	X_{31}	$\overline{S}_1$	$\sum_t (X_{t1} - \overline{S}_1)^2$
2	X_{12}	X_{22}	X_{32}	$\overline{S}_2$	$\sum_t (X_{t2} - \overline{S}_2)^2$
3	X_{13}	X_{23}	X_{33}	$\overline{S}_3$	$\sum_t (X_{t3} - \overline{S}_3)^2$
4	X_{14}	X_{24}	X_{34}	$\overline{S}_4$	$\sum_t (X_{t4} - \overline{S}_4)^2$
Mean	$\overline{T}_1$	$\overline{T}_2$	$\overline{T}_3$		

$$\text{Grand mean } \overline{X} = \frac{\sum_t \sum_s X_{ts}}{mn} \qquad SS_{\text{tot}} = \sum_t \sum_s (X_{ts} - \overline{X})^2$$

x_{ts} t – treatment group $t = 1, \ldots, m$

s – subject number $s = 1, \ldots, n$

1. Subject Means: $\bar{S}_1, \bar{S}_2, \ldots, \bar{S}_n$ $\left(\bar{S}_s = \frac{1}{m} \sum_{t=1}^{m} x_{ts} \right)$

2. Treatment Means: $\bar{T}_1, \bar{T}_2, \ldots, \bar{T}_m$ $\left(\bar{T}_t = \frac{1}{n} \sum_{s=1}^{n} x_{ts} \right)$

3. Grand Mean: $\bar{X} = \frac{1}{mn} \sum_{t=1}^{m} \sum_{s=1}^{n} x_{ts} = \frac{1}{m} \sum_{t=1}^{m} \bar{T}_t = \frac{1}{n} \sum_{s=1}^{n} \bar{S}_s$

4. Total Sum of Squares: $SS_{tot} = \sum_{t=1}^{m} \sum_{s=1}^{n} \left(x_{ts} - \bar{X} \right)^2$ $DF_{tot} = mn - 1$

Individual measurement

Grand mean

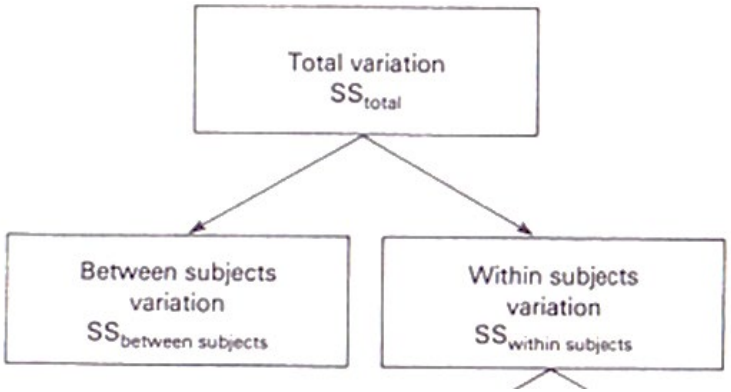
x_{ts} t – treatment group $t = 1, \ldots, m$

s – subject number $s = 1, \ldots, n$

5. Sum of Squares Within Subjects:

$$SS_{wit\,subjs} = \sum_{s=1}^{n} \underbrace{\left[\sum_{t=1}^{m} \left(x_{ts} - \bar{S}_s \right)^2 \right]}_{SS_{subj}} = \sum_{s=1}^{n} SS_s$$

$$DF_{wit\,subjs} = n(m-1)$$



6. Sum of Squares Between Subjects:

$$SS_{bet\,subjs} = m \sum_{s=1}^{n} \left(\bar{S}_s - \bar{X} \right)^2 \qquad DF_{bet\,subjs} = n - 1$$

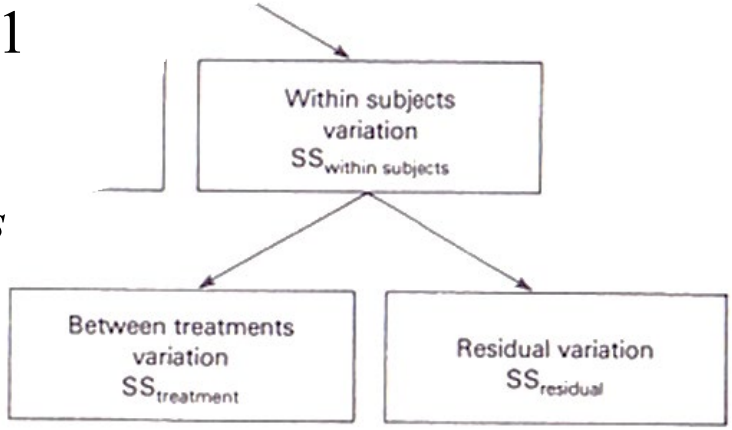
because each subject mean involves m treatments

Process: Separate Within-Subject Sum of Squares into Treatment and Residual Sum of Squares

7. Sum of Squares Due to Treatment:

$$SS_{treat} = n\sum_{t=1}^{m}\left(\bar{T}_t - \bar{X}\right)^2 \qquad DF_{treat} = m-1$$

because each treatment mean involves n subjects



8. Residual Sum of Squares:

$$SS_{wit\,subjs} = SS_{treat} + SS_{res}$$

$$SS_{res} = SS_{wit\,subjs} - SS_{treat} \qquad DF_{res} = (n-1)(m-1)$$

9. Calculate *F* statistic:

$$F = \frac{SS_{treat} / DF_{treat}}{SS_{res} / DF_{res}} = \frac{MS_{treat}}{MS_{res}}$$

$$F_{crit} = F_{\alpha}(DF_{treat}, DF_{res})$$

$$DF_{treat} = m - 1 = \nu_n$$

$$DF_{res} = (n-1)\cdot(m-1) = \nu_d$$

Pros & Cons of Repeated Measures Study Design

- Pros
  - Patient acts as his/her own control
  - Avoids issues of between-subject variability in baseline and response to treatment
  - Need fewer subjects to still have high power of test

- Cons
  - Can be difficult to get same patient back for multiple measurements
  - Requires no carry-over effect between treatments/observation events
  - Disease must be stable and ongoing (i.e., chronic vs. acute)
  - Ethical issues

Note: usage of medical "twins" can get around some of the limitations/cons of repeated measures study designs